

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Software Engineering
FALL SEMESTER, Batch 2021
MIDTERM EXAMINATION

Max.Marks: 20
Date: 11th -01-2022
Roll Number:

Name:

PAPER: A

1. **Elaborate** the purpose of pipeline mechanism in 8086 architecture C-2 (1)
Both units operate asynchronously to give the 8086 an overlapping instruction fetch and execution mechanism which is called as Pipelining. This results in efficient use of the system bus and system performance
2. **Determine** the functions of units of microprocessor C-3 (2)
Bus Interface Unit (BIU) BIU fetches instructions, reads data from memory and I/O ports, writes data to memory and I/O ports. EU decodes and executes instructions that have already been fetched by the BIU. BIU and EU functions separate
3. **Estimate** data bytes that can be processed by the 64bit processor **8 BYTES** C-2 (1)
Give main difference between DDR-SDRAM and SDDRAM C-2 (1)
SDRAM can only send data once per clock. Double-data-rate is Enhanced version of SDRAM .
. Double-data-rate SDRAM can send data twice per clock cycle, Once on the Rising edge and once on the falling edge
4. How can we **differentiate** between STATIC RAM ,DYNAMIC RAM C-2 (2)
Bits stored as charge in capacitors Absence and presence of charges is interpreted as a binary 0 & 1, Need refresh circuits
Simpler construction, Less expensive, Slower
Uses flip-flops Bits stored as on/off switches Binary value stores using flip flop, No refreshing needed
More complex construction, More expensive than DRAM ,Faster than DRAM
5. Cache is Small block of **very fast** temporary memory **defend** this statement with a valid reason C-2 (1)
A cache is a smaller, faster memory, located closer to a processor or on the CPU chip itself and stores copies of the data or instructions that are frequently used.
6. **Determine** the reason why access time of RAM is less than the hard disk C-3 (2)
In RAM Individual addresses identify locations exactly at the same amount of time .Access time is independent of location or previous access
In Hard disk Access time depends on location and previous location Individual blocks have unique address, Access is done by jumping to vicinity plus sequential search
7. **Calculate** the hit ratio and miss ratio if you have 51 cache hits and three misses over a period of time, C-3 (2)
Cache hits =51
Cache misses = 3 **Cache access = 54**
hit ratio= 51/54=0.944 **miss ratio = 1-0.94 =0.06**
8. A computer has a single cache with a 2 ns hit time and a 98% hit rate. Main memory has a 40 ns access time. What is the computer's effective access time? If we add an on-chip cache with a 0.5 ns hit time and a 94% hit rate, what is the computer's effective access time? How much of a speedup does the on-chip cache give the computer? C-3 (4)
 1. $Ht1 + (1-h)t2h2 = (2^{10-9} \times 0.98) + (1-0.98)40^{10-9} \times 1 = 2.76ns$
 2. $Ht1 + (1-h1)t2h2 + (1-h1)(1-h2)h3t3 = (0.5^{10-9} \times 0.94) + (1-0.94) (2^{10-9} \times 0.98) + (1-0.94)(1-0.98)40^{10-9} \times 1 = 0.635ns$
9. **Convert** the following C-2 (4)

Decimal	343.97	Octal	247	Binary	10010101.0111
Binary	101010111.1111	Decimal	167	Decimal	149.4375
Octal	527.7605	Binary	010 100 111	Octal	225.34
Hexa	157.F851			Hexa	CD43.7D
Decimal				Decimal	52547.48828

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PAPER: B

1. Cache Speeds up data transfer **defend** this statement with a valid reason. C-2 (1)
Cache speed up the data transfer as it is smaller in size , located closer to a processor or on the CPU chip itself and stores copies of the data or instructions that are frequently used.
2. **Estimate** the number of address bits 8086 microprocessor can write to or from memory C-2 (1)
20bit address can be written to or from memory
Determine the reason why access time of Cache is less than the hard disk C-3 (1)
In Hard disk Access time depends on location and previous location Individual blocks have unique address , Access is done by jumping to vicinity plus sequential search
Cache uses a Associative access method means memory are content addressable
Access time is independent of location or previous access and stores copies of the data or instructions that are frequently used
3. **Given** size of memory is 128MB how may bits are required to address each location of memory uniquely C-2 (1)
27 bits are required to address each location uniquely
4. **Discuss** the effect on performance if two unit operates asynchronously. C-2 (2)
If two units operates asynchronously it means they both they are independent to each other. This results in efficient use of system performance
5. Estimate data bytes that can be processed by the 32bit processor **4 BYTES** C-2 (1)
6. **Determine** one advantage and disadvantage of Super Computer, Mainframe Computer and Micro Computer C-3 (3)

Supercomputers are the specialized and most powerful computer systems that are designed for specific specialized intense scientific calculations that would be practically impossible for other computers otherwise. including the simulation and modeling of such as climate change and the behavior of molecules; data analysis such as genome sequencing and astronomical observations.	Mainframe computers are built to handle a large variety of tasks are used as a storage for large database and serve as a maximum number of users simultaneously	This is the type of computers that you are most likely to use as personal computers. It is a computer system designed to meet a single user's information processing needs. That includes data and word processing, electronic spreadsheets, and database management systems. Portable computers, such as laptops and handheld computers are also called microcomputers.
Super computers are mostly purpose-built for one or a few specific institutional tasks It requires trained and expert staff needs to be appointed for monitoring and administrating a supercomputer	There is also environmental constraints The temperature of the environment shouldn't be too hot A mainframe computer typically comes with some advanced features. Therefore, it requires more resources than a regular system	A microcomputer which is connected to the internet is vulnerable to numerous security threats. Most of the micro computers are equipped with a low power processors. As a result, their performance is typically lesser than both super computer and a mainframe compute

7. Assume a computer has on-chip and off-chip caches, main memory and virtual memory. Assume the following hit rates and access times: on-chip cache 95%, 1 ns, off-chip cache 99%, 10 ns, and main memory: X%, 50 ns, virtual memory: 100%, 2,500,000 ns. Notice that the on-chip access time is 1 ns. Assume that effective access time is 1.6 ns. What should X be to ensure that EAT? C-3 (4)
Hit ratio = 0.99
8. **Calculate** the hit ratio and miss ratio if you have 51 cache hits and four misses over a period of time, C-3 (2)
Cache hits =51
Cache misses = 4 Cache access = 54
hit ratio= 51/55=0.927 miss ratio := 1-0.92 =0.08
9. Convert the following C-2 (4)

Decimal	443.97	Octal	347	Binary	10010101.0111
Binary	110111011.1111	Decimal	231	Decimal	149.4375
Octal	673.7605	Binary	011 100 111	Octal	225.35
Hexa	1BB.F851			Hexa	BD43.7D
Decimal				Decimal	48451.48828

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